

Logarithm Rules

$$\log(x) + \log(y) = \log(x \times y)$$

$$\log(x) - \log(y) = \log\left(\frac{x}{y}\right)$$

$$\log_a(x^b) = b \times \log_a(x)$$

Remember: $\log(x) = b$ is equivalent to $x = 10^b$

- $\log(10) = 1$
- $\log(1) = 0$

Exponent Rules

$$x^a \times x^b = x^{(a+b)}$$

$$\frac{x^a}{x^b} = x^{(a-b)}$$

$$(x^a)^b = x^{(a \times b)}$$

$$x^a \times y^a = (x \times y)^a$$

$$\frac{x^a}{y^a} = \left(\frac{x}{y}\right)^a$$

$$x^{-n} = \frac{1}{x^n}$$

$$x^0 = 1$$

Conversions

$$1 \text{ lb} = 16 \text{ oz} = 454 \text{ g}$$

$$1 \text{ kg} = 2.2 \text{ lbs}$$

$$1 \text{ inch} = 2.54 \text{ cm}$$

$$1 \text{ foot} = 12 \text{ inches}$$

$$1 \text{ yard} = 3 \text{ feet}$$

$$1 \text{ meter} = 1.1 \text{ yards}$$

$$1 \text{ mile} = 5280 \text{ feet}$$

Percent Change

$$\% \text{ change} = \left(\frac{x_2 - x_1}{x_1}\right) \times 100$$

Remember:

- "of" = multiplication
- "equal to" = equal sign

Example: 30% of 100 is equal to x

$$x = (0.30)(100)$$

Solutions & Dilutions

$$C_1 V_1 = C_2 V_2$$

C = concentration V = volume

Remember: Can be used with any concentration unit (molarity, molality, % concentration)

Combining solutions:

$$C_1 V_1 + C_2 V_2 = C_f V_f$$

Example: 60mL of 40% solution is added to 40mL of 90% solution. What is the concentration of the final solution?

$$0.4(60) + 0.9(40) = C_f(100)$$

Distance & Rate

distance = rate $\times$ time

$$\text{rate (average speed)} = \frac{\text{distance}}{\text{time}}$$

Remember: When two things travel *toward* each other, *add* the two velocities together

Example: Two trains are 135 miles apart. Train 1 is heading east at 50 mph. Train 2 is heading west at 40 mph. How long before the trains cross paths?

$$\text{time} = \frac{135 \text{ miles}}{40 \text{ mph} + 50 \text{ mph}} = \frac{135}{90} = 1.5 \text{ hrs}$$

Compound Interest

$$FV = PV \left(1 + \frac{r}{n}\right)^{nt}$$

FV = future value

t = time

PV = present value

n = amount of times interest

r = interest rate

is compounded per unit t

Example: Interest is compounded *quarterly* (every 3 months), and the length of the investment is *two years* ($n = 4$, $t = 2$)

$$FV = PV \left(1 + \frac{r}{4}\right)^{(4 \times 2)}$$

$$\text{Simplified: } FV = PV(1 + r)^t$$

Remember: Use the simplified formula when the interest is compounded one time per unit "t" ($n = 1$)

Example: Interest is compounded *yearly*, and the length of the investment is *five years* ($n = 1$, $t = 5$)

$$FV = PV(1 + r)^5$$

Data Sets

$\cup$ represents union (all data except duplicates)

$\cap$ represents intersection (only data found in all datasets)

Example: $A = [1, 2, 3]$ $B = [1, 3, 4]$

$$A \cup B = [1, 2, 3, 4]$$

$$A \cap B = [1, 3]$$

Combined Work

$$\frac{1}{t_1} + \frac{1}{t_2} + (\dots) + \frac{1}{t_n} = \frac{1}{t_{\text{total}}}$$

Example: If Tom gets a job done in 4 hours (t_1) and Jerry gets it done in 3 hours (t_2), how many hours does it take to get the job done working together (t_{total})?

$$\frac{1}{4 \text{ hours}} + \frac{1}{3 \text{ hours}} = \frac{1}{t_{\text{total}}}$$

$$\frac{3}{12 \text{ hrs}} + \frac{4}{12 \text{ hrs}} = \frac{7}{12 \text{ hrs}} = \frac{1}{t_{\text{total}}}$$

$$7(t_{\text{total}}) = 1(12 \text{ hrs})$$

$$t_{\text{total}} = \frac{12}{7} = 1.714 \text{ hours}$$

Combinations vs. Permutations

$$\text{combinations} = \frac{n!}{k!(n-k)!} \quad \text{permutations} = \frac{n!}{(n-k)!}$$

n = # of objects to choose from

k = # of chosen objects

Remember:

- Use *combination* when the order *does not* matter. These *are* the same: [ABC], [BCA], [CBA]
- Use *permutation* when the order *does* matter. These *are not* the same: [ABC], [BCA], [CBA]

Dice

Total possible combinations if rolling 2 dice: $6 \times 6 = 36$

Example: What is the probability of rolling 2 dice that add up to the number 4?

3 possible combinations: [2, 2], [1, 3], [3, 1]

$$\text{Therefore, the probability is } \frac{3}{36} = \frac{1}{12}$$

Letter Arrangements

- If all letters are unique (no repeats), the number of possibilities is $n!$
- If there are repeated letters, find the number of possibilities by dividing by the factorials of number of repeats:

$$\frac{n!}{(\text{repeat}_1!)(\text{repeat}_2!)\dots}$$

Example: How many different ways can the letters in the word APPALOOSA be arranged?

Repeats of A: 3

Repeats of O: 2

Repeats of P: 2

Therefore, the number of possibilities is $\frac{9!}{3! \times 2! \times 2!} = 15,120$

Probability

And/or probabilities:

- If *and* is used, multiply probabilities
- If *or* is used, add probabilities

At least one probability: $P(\geq 1) = 1 - P(\text{none})$

- Independent events: same probability each trial; use an exponent for # trials (e.g., $P(\text{none})^n$)
- Dependent events (no replacement): probabilities change each round; multiply probabilities to calculate $P(\text{none})$

Example: What is the probability of at least one boy among 3 children?

$$P(\text{at least one boy}) = 1 - (0.5)^3 = 0.875$$

Deck of Cards

A standard deck has:

- 52 cards (without jokers)
- 4 suits (hearts, clubs, spades, diamonds)
- 13 cards per suit

Remember:

- With replacement:* Return the card after picking it, keeping the number of cards constant
- Without replacement:* Remove the card from the deck after picking it, decreasing the deck size and cards of interest

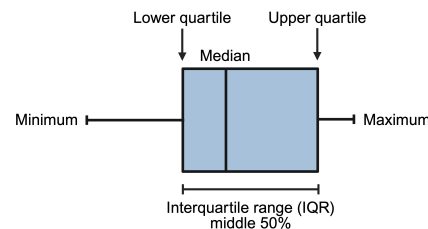
Example: What is the probability of pulling 3 spades in a deck without replacement?

$$\frac{13}{52} \times \frac{12}{51} \times \frac{11}{50} = \frac{11}{850}$$

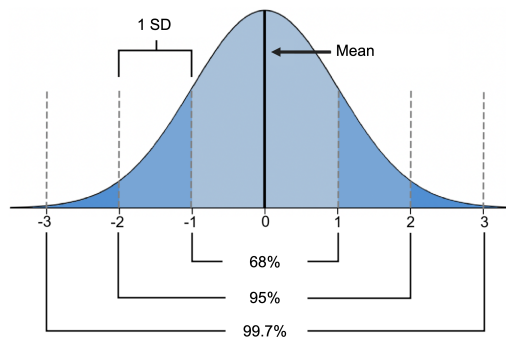
Statistics

Mean	average value ($\bar{x}$) = $\frac{\text{sum of data points}}{\text{\# of data points}}$	
Median	middle number of a dataset (order values numerically)	
Mode	most frequently occurring number in dataset	
Standard deviation (σ)	measure of how spread out the values of a dataset are $\sigma = \sqrt{\frac{\sum(x - \bar{x})^2}{N}}$	x = each value $\bar{x}$ = mean N = population size
Variance	average of squared deviations from the mean variance = σ^2	
Z score	number of standard deviations a value is from the mean (e.g., +2.5 z-score represents a value that is 2.5 SD away from the mean) $\text{z-score} = \frac{(x - \mu)}{\sigma}$	x = datapoint μ = mean σ = standard deviation

Box and Whisker Plot



Normal Distribution



In normal distribution:

- 68% of data is within 1 standard deviation of the mean
- 95% of data is within 2 standard deviations of the mean
- 99.7% of data is within 3 standard deviations of the mean

Geometry

Circle	Area = πr^2 Circumference = πd or $2\pi r$	r = radius d = diameter
Triangle	Area = $\frac{1}{2}bh$	b = base h = height
Square	Area = s^2	s = side length

Linear Equations

Line formulas	$y = mx + b$ $m = \frac{(y_2 - y_1)}{(x_2 - x_1)}$	m = slope b = y-intercept
Distance between 2 points	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	
Midpoint of a line	$\left[\left(\frac{x_1 + x_2}{2} \right), \left(\frac{y_1 + y_2}{2} \right) \right]$	

Remember:

- In *parallel* lines, slopes are *equivalent* (e.g., $m = 2$ and $m = 2$)
- In *perpendicular* lines, slopes are *negative reciprocals* (e.g., $m = -4$ and $m = \frac{1}{4}$)

Quadratic Equations

For any equation in the format $ax^2 + bx + c = 0$

- $(x + y)^2 = x^2 + 2xy + y^2$
- $(x - y)^2 = x^2 - 2xy + y^2$
- $(x + y)(x - y) = x^2 - y^2$

$$\text{Quadratic formula } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\text{Parabola vertex x-coordinate} = -\frac{b}{2a}$$

Remember: If $(b^2 - 4ac)$ is:

- positive, there are *2 real solutions*
- 0, there is *1 real solution*
- negative, there are *0 real solutions*